

CLUE-AI: A Collaborative Game-Based Learning Approach to Promote Critical Literacy for Uncovering Errors in AI

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Abstract. As generative AI tools become embedded in educational practice, students must develop the skills to critically evaluate AI outputs. However, little empirical work has examined whether they can recognize and avoid incorporating problematic AI responses into their own work. We report results from a pilot study ($N = 30$) investigating a collaborative digital game designed to develop critical AI literacy by teaching students to distinguish facts from AI misrepresentations and hallucinations. After gameplay, participants’ distinguishing ability improved significantly across repeated gameplay cycles (55.9% to 70.6%), and participants’ conceptual definitions of both hallucination and misrepresentation improved significantly from pre to post. Low-performing students benefited most from collaboration, showing significant learning gains following paired discussions. These findings suggest that situated, collaborative game-based interventions can shift students from passive consumers of AI output toward critical evaluators.

Keywords: Critical AI Literacy · Collaborative Learning · Digital Game-Based Learning · AI Hallucination · AI Misrepresentation

1 Introduction

Generative AI (GenAI) tools are rapidly becoming embedded in students’ everyday educational practices, supporting activities such as writing, coding, brainstorming, and information summarization. As students increasingly interact with AI—often without explicit guidance—it becomes critical not only to consider how these tools are used, but how their outputs are interpreted and evaluated. Prior work shows that users tend to over-rely on AI systems, frequently deferring to AI-generated responses in ways that reduce critical engagement and independent reasoning [13]. In post-secondary education, this pattern is particularly concerning as students may prioritize speed and convenience over reflection, increasing the risk of incorporating biased, misleading, or fabricated information into their academic work.

Addressing this gap requires shifting focus from discouraging AI use toward developing students’ *critical AI literacy*, which we define as the capacity to identify misrepresented or fabricated information, interrogate sources, and maintain epistemic agency when working with AI. Drawing from traditions of critical literacy [4], recent frameworks argue that students must be equipped not only to use AI tools but to question, critique, and reflect on how these systems shape knowledge, power, and representation [11]. A central component of this literacy is understanding forms in which AI inaccuracies occur. Two of the most common AI failures we focus on, in this paper, are AI hallucinations and misrepresentations. We define *AI hallucinations* as outputs where AI generates a convincing, contextually coherent but entirely fabricated response independent of factual basis [9]. We define *AI misrepresentation* as systematic distortions embedded within AI outputs that can reinforce existing biases and amplify prejudice and stereotype [10]. For students, uncritical exposure to such outputs risks normalizing biased perspectives as objective truth. Developing critical AI literacy is therefore essential to equip students to engage with GenAI reflectively, recognizing both its utility and its impediments.

Constructivist learning theory, commonly adopted in AI literacy education, emphasizes that learners actively construct knowledge through experience and social interaction [3]. Collaborative learning offers a natural pedagogical foundation for developing critical AI literacy by helping students construct knowledge through dialog, negotiation, and shared sense-making rather than passive reception of information [2]. This is important when engaging with complex AI-related phenomena such as misrepresentations and hallucinations, which can be difficult to identify or conceptualize individually. Prior work also suggests that collaborative settings support deeper critical engagement with sociotechnical systems by making uncertainty, assumptions, and interpretive gaps visible [8]. We therefore hypothesized that discussing AI output in pairs can allow students to surface disagreements, compare interpretations, and collectively interrogate why an AI response may be misleading or biased.

Digital game-based learning (DGBL) complements collaborative learning by situating practice in meaningful, low-stakes environments that motivate engagement with complex concepts [5]. Research has demonstrated that well-designed games can match or exceed traditional instruction across domains such as STEM education, [12], and AI Education [14], among others. DGBL environments engage learners in iterative processes of exploration and rule construction. These processes align with active, constructive, and interactive forms of learning that support the refinement of mental models over time [1]. This makes DGBL especially appropriate for learning goals that involve interpretation, judgment, and conceptual change. When combined with collaborative mechanisms that support shared sense-making, we hypothesize that DGBL can support deep learning by encouraging learners to articulate their reasoning, attend to alternative perspectives, and negotiate meaning with peers—all essential for developing both critical dispositions toward AI and understanding of AI limitations.

In this paper, we present CLUE-AI (**C**ritical **L**iteracy for **U**ncovering **E**rrors in **AI**), a collaborative digital game designed as a proof-of-concept instructional intervention that combines DGBL and collaborative learning to develop critical AI literacy among university students. The game teaches students to distinguish factual information from AI misrepresentations and hallucinations through repeated cycles of individual classification and paired deliberation (as explained in 2. We report results from a pilot study attempting to assess the game’s effectiveness by looking at learning gains in conceptual understanding of AI misrepresentation and hallucination, in-game performance across play cycles, and the role of collaboration in supporting individual learning.

2 CLUE-AI System Design

CLUE-AI builds on the narrative foundation of the game Case 429, where players step into the role of Watson, co-detective to Sherlock Holmes, to solve the murder of Sir Eric Harpe [7]. Scotland Yard’s AI tool, framed as trained on thousands of criminal cases, provides players with AI-generated summaries of case files for each suspect. Each suspect’s file contains long, reading-heavy documents such as interrogation transcripts and background information. Working under a time constraint, players must treat the AI summaries as investigative aids while simultaneously evaluating their accuracy against original source texts. Players classify each AI statement as **Accurate**, **Misrepresentation**, or **Hallucination** before deciding whether to **Accuse** the suspect or rule them out as **Inconclusive**. The practice of classification is primarily introduced to prime players to be critical of AI and not to defer their judgment to machines. Following Veldhuis et al.’s [11] critical AI literacy framework, the classification task in the game also addresses the foundational learning objective of *"question preconceptions of AI and its critical implications"* by revealing how AI systems can misrepresent information through bias, and hallucinate to fabricate non-existent information. Additionally, players must cross-reference source evidence during classification, which reinforces the skill of critically evaluating AI outputs. Expanding this design to include collaboration allows players to take a step further in building a deeper understanding of the concepts of AI misrepresentation and hallucination. The game has players consider each other’s viewpoints, meeting the learning objectives of *"making difference visible and reflecting on contradictory perspectives"* from Veldhuis et al. [11].

As shown in Fig. 1, the game is scaffolded across multiple phases. Players first spend 8 minutes individually classifying AI statements, ensuring both partners develop their own interpretations before negotiation begins. They then deliberate for 8 minutes to resolve conflicts and submit a single, unified classification sheet. During this collaboration phase, the system automatically highlights divergent classifications, directing attention to productive disagreements and requiring players to defend their reasoning with evidence. Unlike voting or averaging, forced consensus keeps both players cognitively engaged throughout. Submitting the joint sheet reveals the correct classification key, putting players

in the feedback phase where they can choose to spend as much or as little as they like before selecting the next suspect to investigate. Players experience at most 3 cycles of the game flow depending on whether they accuse a suspect or not. Players also encounter variable levels of difficulty depending on the suspect they investigate.

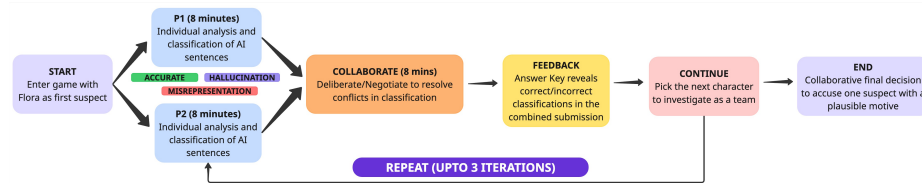


Fig. 1. Game Flow

The Wizard-of-Oz AI summaries were written by three human researchers. Accurate sentences have direct references in source documents. Misrepresentation sentences embed subtle intersectional biases, for example, overemphasizing criminal suspicion for characters from marginalized groups while dismissing incriminating financial evidence for a wealthier suspect. Hallucination sentences either alter real quotes or fabricate claims with no source support. The game runs on a modular web architecture with a browser-based client, backend API, and cloud datastore; it does not capture audio, video, or player-to-player chat.

3 Methodology

We recruited 30 participants (8 males, 22 females; 23 undergraduates, 7 graduate students) from a Midwestern University for a within-subjects study conducted via Zoom under IRB approval. Participants were randomly paired into groups of two. All participants self-reported using Generative AI tools like ChatGPT, Gemini, or Claude to some capacity in their daily lives.

Participants completed an individual pretest followed by 45-60 minutes of gameplay, and an individual post-test, for a total session of 90 minutes. The tests had 2 short-answer where participants defined "misrepresentation" and "hallucination", and 2 multiple-choice questions (MCQs) on AI limitations (from the Generative AI Literacy Assessment Test [6]). Before beginning gameplay, participants were instructed not to use any external search engines or AI tools and to rely only on each other throughout the game especially the collaborative phase. For collaborative segments, participants were informed that screens would be synchronized and that selections could affect their partner's screen. We used a mixed-method approach to analyze the game interaction logs, pre- and post-test responses, and transcripts of participant deliberations.

4 Results

Our results include data from 15 participant pairs ($N = 30$), out of which 7 successfully identified the correct murderer and the motive. Out of these 7, one

pair identified the correct murderer and motive after 2 attempts, therefore they only investigated 2 characters, while all other participants investigated 3 characters (i.e., received 3 attempts at learning by participating in the classification of AI statements). Present below are results indicating participants' in-game performance and understanding of AI misrepresentation & hallucination.

4.1 Did participants' understanding of AI misrepresentations and hallucinations change pre to post?

To compare participants' understanding of AI misrepresentation and hallucinations, we asked them to define the terms in their own words. Three independent raters scored the accuracy of participants' definitions on a scale of 0-3 using the coding scheme in Table 1. The Inter-rater reliability assessed using Fleiss' kappa yielded moderate agreement across all measures: Pre-test misrepresentation ($\kappa = .55$), post-test misrepresentation ($\kappa = .41$), pre-test hallucination ($\kappa = .46$), and post-test hallucination ($\kappa = .55$). The overall average reliability was moderate ($\kappa = .49$).

Table 1. Scoring Rubric for AI Hallucination and Misrepresentation

Score	AI Misrepresentation	AI Hallucination
3	Mentions bias, stereotyping, or skewed interpretation; incorrect interpretation of existing information; partial truths or context removal. <i>Example:</i> "AI misrepresentation is when AI uses biased training data to produce skewed interpretations"	Mentions fabrication/making up/invention of information; lack of factual basis or source; information is false/non-existent. <i>Example:</i> "AI hallucination is when AI fabricates information without factual support"
2	Mentions incorrect information but unclear if it's about interpretation vs. fabrication; vague mention of bias without explanation. <i>Example:</i> "AI gives biased answers"	Captures ONE core element (either fabrication OR lack of source); may confuse with general errors/mistakes. <i>Example:</i> "When AI gives wrong information"
1	Very general or conflates with hallucination; no clear understanding of bias/interpretation issues. <i>Example:</i> "AI makes mistakes"	Very general or conflates with other concepts; no mention of fabrication or source issues. <i>Example:</i> "AI gets confused" or "AI misunderstands things"
0	"I don't know"; empty response or "-"	"I don't know"; empty response or "-"

Participants demonstrated improvement in their understanding of AI misrepresentation from pretest ($M = 1.87, SD = 1.11$) to post-test ($M = 2.43, SD = .73$), $t(29) = 2.29, p = .030, d = .42$. From explaining AI misrepresentations as "AI may generate wrong or fake information (P19)", participants' explanations evolved to "AI may retrieve **biased** information and draw erroneous conclusions, thereby misleading users (P19)". Similarly, participants' understanding

of AI hallucination improved significantly from pre-test ($M = 1.60, SD = .86$) to post-test ($M = 2.17, SD = .83$), $t(29) = 3.46, p < .001, d = .63$. Participants went from describing AI Hallucinations as "*when ai makes up data or anything (P01)*" to more nuanced definitions that included both fabrication of information and lack of factual sources like "*AI makes up information based on nonexistent sources (P29)*". These findings suggest that having conversations around these abstract concepts successfully enhanced participants' conceptual understanding of both AI phenomena, hallucination more than misrepresentation given the magnitude of improvement, while providing them the appropriate contextual vocabulary to explain their understanding.

4.2 Did participants learn to differentiate between accurate information, AI misrepresentations, and hallucinations in game?

We analyzed participants' accuracy in classifying AI statements using a linear mixed-effects regression model while accounting for individual differences. Our results suggest that classification accuracy increased significantly with each character investigated ($\beta = .073, SE = .021, p < .001$), improving mean accuracy from 55.9% on the first character to 70.6% on the third character (see bar graphs in Fig. 2). Neither pretest scores on the MCQs ($\beta = -.053, p = .178$) nor correct identification of the murder ($\beta = -.021, p = .583$) predicted in-game classification accuracy.

However, classification accuracy varied substantially by sentence type, $F(2, 81) = 58.30, p < .001$. Participants were most accurate at identifying *accurate* sentences ($M = 81.9\%, SD = 9.9\%$) compared to *hallucinations* ($M = 46.3\%, SD = 18.9\%$) and *misrepresentations* ($M = 39.1\%, SD = 17.7\%$). These differences emerged through distinct learning trajectories (see line plots in Fig. 2) While accuracy on accurate sentences improved from 73.4% to 87.2% across character investigations, and hallucination detection improved modestly from 42.9% to 52.7%, misrepresentation detection declined from 46.6% to 27.8%. This pattern suggests that while students learned to recognize truthful information, they grew increasingly susceptible to subtle factual distortions.

4.3 Did collaboration support learning?

To examine whether collaborative discussion facilitated individual learning, we analyzed 57 character-to-character transitions³ from 30 participants, testing whether collaborative benefit (group accuracy minus individual accuracy) predicted their subsequent individual performance on the next character. To avoid confounding between collaborative benefit and baseline performance, we classified transitions as low-performing (LP) ($n = 30$) or high-performing (HP) ($n = 27$) based on whether the student's baseline accuracy on the previous character

³ One participant pair completed only two characters rather than three, due to early identification of the murderer and one participant from a different pair had a technical glitch.

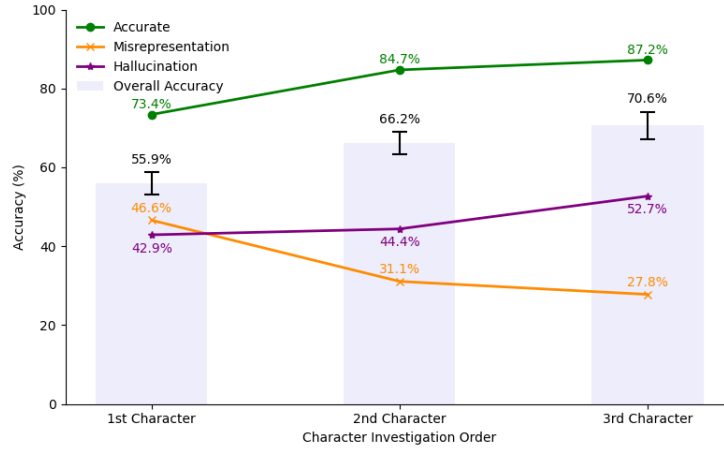


Fig. 2. Differences in accuracy across characters representing learning

was below or above the median. For LP participants, collaborative benefit significantly predicted learning gains ($r = .40, p = .030$); for HP participants, no relationship was observed ($r = -.01, p = .961$). LP participants showed positive learning gains ($M = 19.6\%$) while HP participants showed slight declines ($M = -5.2\%$), suggesting that knowledge asymmetry helped LP students. Evidence for how this collaborative learning occurred can be seen in the conversation transcripts from pairs where LP participants showed substantial improvement. These conversations reveal several mechanisms through which HP students scaffolded their partners' understanding:

Sharing individual working definitions. Some students articulated their understanding of the definitions: *"I interpret hallucination to mean the AI is saying something that is just flat-out not true. It's just making something up that's totally not based on any facts"* (Pair 15, investigating Flora).

Explanation and reasoning. When one participant was uncertain, their partner often provided explicit reasoning. For example, *"LP participant: I am confused [about] the last two sentences..."*; *"HP Participant: So the AI is claiming that as a black stepchild of a white father, the subject may have had a troubled childhood. The right answer is a hallucination, because, I guess, the AI is kind of just saying the subject definitely, or probably could have had a troubled childhood. Just because of this one thing, AI imagines the childhood part."* (Pair 9, analyzing the answer key).

Error correction. HP participants gently corrected misclassifications: *"Wait, I see why you had that accurate because it's pulling a direct quote but she never said I'd kill him...she said I'll burn his house down, which is why it's Hallucination, because it came up with its own inaccurate thing"* (Pair 6, investigating Flora).

These conversational patterns suggest that collaboration could have supported shared-knowledge building, with HP participants providing explanations and reasoning that LP participants could utilize in subsequent individual classifications.

5 Discussion & Future Directions

Our findings suggest that CLUE-AI helped improve participants' understanding of AI misrepresentation and hallucinations. Across the multiple game cycles, their overall classification accuracy improved significantly as well. However, while participants improved at recognizing accurate information and hallucinations, their ability to detect misrepresentations actually declined from 46.6% to 27.8%. One possible explanation for this is that as students learned to verify factual claims against sources, they grew overconfident in plausible-sounding statements that contained subtle bias rather than outright fabrication. If confirmed in a study with a larger sample size, this would suggest that misrepresentation detection requires different pedagogical scaffolding than hallucination detection. A follow-up study could also test this by comparing conditions with and without targeted misrepresentation training.

The collaborative learning findings offer another direction for future work. We found that only low-performing students benefited significantly from collaborative discussion (see Section 4.3). However, our analysis measured whether collaboration predicted subsequent individual improvement, but did not examine the dynamics of the collaboration itself, such as which partner dominated the conversation or whether consensus reflected genuine negotiation versus conformity to a more assertive partner. What appears as learning gains for low-performers could instead reflect confirmation bias during collaboration that happened to align with correct answers. Our future work will analyze turn-taking patterns, classification change sequences, and argument structure in the deliberation transcripts to distinguish productive knowledge co-construction from superficial agreement.

It is important to note that this was a pilot study with 30 participants making it underpowered to detect smaller effects or to examine nuanced individual differences. We did not account for variability in difficulty across characters. While Flora was the mandatory first investigation for all pairs, participants chose their second and third characters from four options, introducing uncontrolled variance in task difficulty across learning trajectories. Additionally, our quantitative analyses focused on timed classification phases; we did not systematically analyze conversational data from untimed segments such as answer key review and character selection, which may contain rich evidence of metacognitive reasoning. Finally, we did not measure participants' prior experience with AI tools beyond confirming general use; future powered studies should assess baseline AI literacy and usage frequency to examine whether intervention effects vary by prior exposure.

In conclusion, this pilot study demonstrates that a single session interaction with CLUE-AI can produce measurable shifts in how students' conceptual understanding of AI's limitations, and boost fact determination ability through the practice of source-verification. These findings suggest that well-designed, situated interventions can rapidly recalibrate students' epistemic stance toward AI, offering a scalable path toward critical AI literacy.

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